

PROGRAM 10:

Write a shell script to print the multiplication table of the given argument using for loop.

AIM

To write a shell script that reads a table number and range from the user and prints its multiplication table using a for loop.

PROCEDURE

STEP 1: Create a file using vi editor.

STEP 2: Read the table number n from the user using the read command.

STEP 3: Read the range from the user using the read command.

STEP 4: Use a for loop starting from 1 up to range. Inside the loop calculate the product using expr and display each row.

STEP 5: Stop the process.

Program:

```
echo "Enter a number"
```

```
read a
```

```
for i in {1..10}
```

```
do
```

```
echo "$i*$a=$((($i*$a))"
```

```
done
```

PROGRAM 11:

Write a shell script to display and analyze group information.

- a. Display all groups available in the system.
- b. Display the group ID of a specified group.
- c. List users belonging to a particular group.
- d. Count the total number of groups available.
- e. Display groups assigned to the current user.

AIM

To write a shell script that displays and analyzes group information: all groups, group ID, group members, total group count, and current user's groups.

PROCEDURE

STEP 1: Create a shell script file using the vi editor.

STEP 2: Use 'cat /etc/group | cut -d: -f1' to display all groups available in the system.

STEP 3: Accept a group name from the user and use 'getent group' to display its group ID.

STEP 4: Accept a group name and use 'getent group' to list all users belonging to that group.

STEP 5: Use 'cat /etc/group | wc -l' to count the total number of groups.

STEP 6: Use the 'groups' command to display all groups assigned to the current user.

STEP 7: Stop the process.

Program:

```
echo "Display all groups available in the system"
cat /etc/group
echo "Display the group ID of a specified group"
getent group it30 | cut -d : -f3
echo "List users belonging to a particular group"
getent group it30 | cut -d : -f4
echo "Count the total number of groups available"
wc -l < /etc/group
echo "Display groups assigned to the current user"
groups
```

PROGRAM 12:

Write a shell script to monitor process activities.

- a. Display all running processes.
- b. Display processes belonging to the current user.
- c. Show top five CPU consuming processes.
- d. Display process ID and parent process ID.
- e. Display the total number of running processes.

AIM

To write a shell script that monitors process activities: all processes, user processes, top CPU consumers, PID/PPID details, and total process count.

PROCEDURE

STEP 1: Create a shell script file using the vi editor.

STEP 2: Use 'ps -ef' to display all running processes in the system.

STEP 3: Use 'ps -u \$USER' to display processes belonging to the current user only.

STEP 4: Use 'ps -eo pid,comm,%cpu --sort=-%cpu | head -6' to show top 5 CPU consuming processes.

STEP 5: Use 'ps -eo pid,ppid,comm' to display process ID and parent process ID.

STEP 6: Use 'ps -ef | wc -l' to count total number of running processes.

STEP 7: Stop the process.

Program:

```
echo "Display all running process"
ps -e
echo "Display process belonging to the current user"
ps -u $(whoami)
echo "Show top five CPU consuming process"
ps -eo pid,comm,%cpu-sort=-%cpu|head -6
echo "Display process ID and parent process ID"
ps -eo pid,ppid,comm
echo "Display the total number of running process"
ps -e h |wc -l
```

PROGRAM 13:

Write a shell script to display network configuration information.

- a. Display system hostname.
- b. Display IP address of all network interfaces.
- c. Display routing table information.
- d. Display DNS server configuration.
- e. Test network connectivity with a remote host.

AIM

To write a shell script that displays network configuration details: hostname, IP addresses, routing table, DNS servers, and tests network connectivity.

PROCEDURE

STEP 1: Create a shell script file using the vi editor.

STEP 2: Use 'hostname' command to display the system hostname.

STEP 3: Use 'ip addr show' to display IP addresses of all network interfaces.

STEP 4: Use 'ip route' to display the routing table information.

STEP 5: Use 'cat /etc/resolv.conf' to display DNS server configuration.

STEP 6: Accept a remote host from the user and use 'ping -c 4' to test connectivity.

STEP 7: Stop the process.

Program:

```
echo "Display system hostname"
hostname
echo "Display IP address of all network interfaces"
ip addr show
echo "Display routing table information"
ip route
echo "Display DNS server configuration"
cat /etc/resolv.conf
echo "Test network connectivity with remote host"
ping -c 4 google.com
```

Program 14:

Write a shell script to monitor system logs and detect suspicious activity.

- a. Display recent system log entries.
- b. Display login history of users.
- c. Display failed login attempts.
- d. Search for specific keywords in log files.
- e. Display the last 10 security related log messages.

AIM

To write a shell script that monitors system logs to detect suspicious activity: recent logs, login history, failed logins, keyword search, and security messages.

PROCEDURE

STEP 1: Create a shell script file using the vi editor.

STEP 2: Use 'tail -20 /var/log/messages' or 'journalctl -n 20' to display recent log entries.

STEP 3: Use 'last | head -15' to display login history of users.

STEP 4: Use grep to search for 'Failed password' in /var/log/secure to find failed login attempts.

STEP 5: Accept a keyword from the user and search for it in log files using grep.

STEP 6: Use 'tail -10 /var/log/secure' to display the last 10 security-related log messages.

STEP 7: Stop the process.

Program :

```
touch ~/mylog.log
```

```
echo "$(date)-INFO-User Logged In">>~/mylog.log
```

```
echo "$(date)-INFO-User Opened Application">>~/mylog.log
```

```
echo "$(date)-WARNING-Incorrect Password Entered">>~/mylog.log
```

```
echo "$(date)-INFO-User Logged OUT">>~/mylog.log
```

```
tail -f ~/mylog.log
```

```
last | tail -10
```

```
grep WARNING mylog.log
```

```
grep INFO mylog.log
```

PROGRAM 15:

Write a shell script to analyze disk usage.

- a. Display disk usage of all directories in the home folder.
- b. Display the top 5 largest directories in the system.
- c. Display the number of files in each directory.
- d. Display the file system type of each partition.
- e. Display free disk space available.

AIM

To write a shell script that analyzes disk usage: home directory sizes, top 5 largest directories, file counts, filesystem types, and free disk space.

PROCEDURE

STEP 1: Create a shell script file using the vi editor.

STEP 2: Use 'du -sh ~/*' to display disk usage of all directories inside the home folder.

STEP 3: Use 'du -sh /* | sort -rh | head -5' to find the top 5 largest directories in the system.

STEP 4: Use a for loop with 'ls | wc -l' inside each subdirectory to count the number of files.

STEP 5: Use 'df -T' to display the filesystem type of each mounted partition.

STEP 6: Use 'df -h' to display free disk space available on all partitions.

STEP 7: Stop the process.

Program:

```
echo "--Disk usage of Home Directory--"
du -sh ~/* 2>/dev/null

echo " "
echo "---Top 5 largest Directories--"
du -sh /* 2>/dev/null | sort -rh | head -5

echo " "
echo "--File count in Each Home Subdirectory--"
for dir in ~/*/
do
count=$(ls "$dir" 2>/dev/null | wc -l)
echo "$dir:$count files"
done

echo " "
echo "--File System Type of each partition--"
df -T | awk '{print $1,$2,$NF}'

echo " "
echo "--- Free Disk Space Available ---"
df -h
```